

Assignment 1 - Understanding Byte Pair Encoding (BPE)

Roll No. 22 | Corpus: read / reading / reader / reads

1. Training Corpus

For Roll No. 22, the last digit is 2. Therefore, the assigned artificial training corpus is: **read (5)**, **reading (3)**, **reader (2)**, and **reads (2)**. The number beside each word is its corpus frequency and is not a token.

Word	Frequency
read	5
reading	3
reader	2
reads	2

2. Initial Symbol Representation

Each word is split into individual characters and the end-of-word marker _ is appended.

5 r e a d _

3 r e a d i n g _

2 r e a d e r _

2 r e a d s _

Pair frequencies are weighted by the word frequency. For example, the pair **r e** occurs once in every listed word, so its weighted frequency is $5 + 3 + 2 + 2 = 12$.

Tie-breaking rule: if multiple pairs have the same maximum frequency, choose the pair encountered first when scanning the words in the listed order and from left to right.

3. Iteration 1

Initial weighted pair counts:

Adjacent pair	Weighted frequency
r e	12
e a	12
a d	12
d _	5
d i	3
i n	3
n g	3
g _	3
d e	2
e r	2
r _	2
d s	2

Adjacent pair	Weighted frequency
s _	2

The maximum frequency is **12**, shared by **r e**, **e a**, and **a d**. By the tie-breaking rule, **r e** is encountered first.

Assignment 1 - Understanding Byte Pair Encoding (BPE)

Roll No. 22

Merge: r + e -> re

Updated corpus:

5 re a d _

3 re a d i n g _

2 re a d e r _

2 re a d s _

4. Iteration 2

The adjacent pairs are counted again after the first merge:

Adjacent pair	Weighted frequency
re a	12
a d	12
d _	5
d i	3
i n	3
n g	3
g _	3
d e	2
e r	2
r _	2
d s	2
s _	2

The maximum frequency is again **12**, shared by **re a** and **a d**. The pair **re a** occurs first under the stated rule.

Merge: re + a -> rea

Updated corpus:

5 rea d _

3 rea d i n g _

2 rea d e r _

2 rea d s _

5. Iteration 3

Recalculate the pair frequencies:

Adjacent pair	Weighted frequency
rea d	12

Adjacent pair	Weighted frequency
d _	5
d i	3
i n	3
n g	3
g _	3
d e	2
e r	2
r _	2
d s	2
s _	2

The pair **rea d** has the highest frequency, **12**. Therefore it is selected for the third merge.

Assignment 1 - Understanding Byte Pair Encoding (BPE)

Roll No. 22

6. Third Merge

Merge: rea + d -> **read**

Updated corpus after three merges:

5 read _
3 read i n g _
2 read e r _
2 read s _

7. Summary of the Three BPE Merges

Iteration	Most frequent pair	Weighted frequency	New token
1	r e	12	re
2	re a	12	rea
3	rea d	12	read

Final learned merge sequence: r e -> re, re a -> rea, rea d -> read.

After the three merges, the common stem **read** has been learned as a subword unit. The remaining suffix characters in *reading*, *reader*, and *reads* remain separate because only three merges were requested.

8. Final Result

Word frequency	Final representation after 3 merges
5	read _
3	read i n g _
2	read e r _
2	read s _

The word frequencies were used only as weights while counting adjacent pairs. At every iteration, the pair counts were recalculated from the updated corpus before selecting the next merge.